

Commerce XI Maths App

Theory (Pages 31-33)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 32

Question 1 - Theory P31

Combined data means data obtained by

Options

- A. joining two or more data sets
- B. removing all frequencies
- C. using only quartiles
- D. drawing an ogive

Correct answer: joining two or more data sets

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives joining two or more data sets.

Question 2 - Theory P31

For samples of sizes n_1 and n_2 , combined size is

Options

- A. $n_1 + n_2$
- B. $n_1 \times n_2$
- C. $n_1 - n_2$
- D. $(n_1 + n_2)/(2)$

Correct answer: $n_1 + n_2$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $n_1 + n_2$.

Question 3 - Theory P31

The combined mean is

Options

- A. $(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$
- B. $(\bar{x}_1 + \bar{x}_2)/(2)$ always
- C. $n_1\bar{x}_1 - n_2\bar{x}_2$
- D. $\sigma_1 + \sigma_2$

Correct answer: $(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$.

Question 4 - Theory P31

In the combined S.D. formula, d_1 equals

Options

A. $\bar{x}_1 - \bar{x}_c$

B. $\bar{x}_c - \bar{x}_2$

C. $\sigma_1 - \sigma_c$

D. $n_1 - n_2$

Correct answer: $\bar{x}_1 - \bar{x}_c$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\bar{x}_1 - \bar{x}_c$.

Question 5 - Theory P31

In the combined S.D. formula, d_2 equals

Options

A. $\bar{x}_2 - \bar{x}_c$

B. $\bar{x}_c - \bar{x}_1$

C. $\sigma_2 - \sigma_c$

D. $n_2 - n_1$

Correct answer: $\bar{x}_2 - \bar{x}_c$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\bar{x}_2 - \bar{x}_c$.

Question 6 - Theory P31

Combined variance includes both within-group variance and

Options

A. differences between group and combined means

B. class limits only

C. quartiles only

D. ranges only

Correct answer: differences between group and combined means

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives differences between group and combined means.

Question 7 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45, the combined mean is

Options

- A. 38
- B. 34.5
- C. 35
- D. 69

Correct answer: 38

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 38.

Question 8 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and combined mean 38, d_1 is

Options

- A. -14
- B. 14
- C. -7
- D. 21

Correct answer: -14

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives -14.

Question 9 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and combined mean 38, d_2 is

Options

- A. 7
- B. -7
- C. 14
- D. 21

Correct answer: 7

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 7.

Question 10 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and standard deviations 6 and 11, the combined variance is approximately

Options

- A. 190.67
- B. 13.8
- C. 121
- D. 36

Correct answer: 190.67

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 190.67.

Question 11 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and standard deviations 6 and 11, the combined standard deviation is approximately

Options

- A. 13.8
- B. 190.67
- C. 17
- D. 8.5

Correct answer: 13.8

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 13.8.

Question 12 - Solved Example P31-32

If the first group has 100 items and the combined group has 250, the second group contains

Options

- A. 150 items
- B. 100 items
- C. 250 items
- D. 350 items

Correct answer: 150 items

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 150 items.

Question 13 - Solved Example P31-32

	First group	Combined group
Number of items	100	250
Mean	45	51
Variance	49	130

The mean of the second group is

Options

A. 55

B. 51

C. 45

D. 60

Correct answer: 55

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 55.

Question 14 - Solved Example P31-32

	First group	Combined group
Number of items	100	250
Mean	45	51
Variance	49	130

The variance of the second group is

Options

A. 144

B. 12

C. 130

D. 49

Correct answer: 144

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 144.

Question 15 - Solved Example P31-32

	First group	Combined group
Number of items	100	250
Mean	45	51
Variance	49	130

The standard deviation of the second group is

Options

- A. 12
- B. 144
- C. 11
- D. 14

Correct answer: 12

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 12.

Question 16 - Theory P32

Standard deviation depends on the

Options

- A. unit of measurement
- B. order of observations only
- C. name of the variable
- D. sample label

Correct answer: unit of measurement

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives unit of measurement.

Question 17 - Theory P32

Coefficient of variation is independent of the

Options

- A. unit of measurement
- B. mean
- C. standard deviation
- D. sample size

Correct answer: unit of measurement

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives unit of measurement.

Question 18 - Theory P32

Coefficient of variation is calculated as

Options

- A. $100 \times (\sigma)/(\bar{x})$
- B. $100 \times (\bar{x})/(\sigma)$
- C. $(\sigma^2)/(\bar{x})$
- D. $\bar{x} - \sigma$

Correct answer: $100 \times (\sigma)/(\bar{x})$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $100 \times (\sigma)/(\bar{x})$.

Question 19 - Theory P32

Coefficient of variation is expressed as a

Options

- A. percentage
- B. class interval
- C. frequency
- D. raw score

Correct answer: percentage

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives percentage.

Question 20 - Theory P32

A distribution with the smaller C.V. is more

Options

- A. consistent
- B. variable
- C. skewed
- D. dispersed

Correct answer: consistent

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives consistent.

Question 21 - Theory P32

A distribution with the larger C.V. is more

Options

- A. variable
- B. consistent
- C. compact
- D. homogeneous

Correct answer: variable

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives variable.

Question 22 - Theory P32

C.V. is particularly useful for comparing data measured in

Options

- A. different units or with different means
- B. the same order only
- C. equal frequencies only
- D. one class only

Correct answer: different units or with different means

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives different units or with different means.

Question 23 - Solved Example P32

Varad has mean 50 and S.D. 11. His C.V. is

Options

- A. 22%
- B. 11%
- C. 25%
- D. 50%

Correct answer: 22%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 22%.

Question 24 - Solved Example P32

Viraj has mean 58 and S.D. 16. His C.V. is approximately

Options

- A. 27.59%
- B. 16%
- C. 22%
- D. 36%

Correct answer: 27.59%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 27.59%.

Question 25 - Solved Example P32

Akhilesh has mean 21 and S.D. 5. His C.V. is approximately

Options

- A. 23.81%
- B. 5%
- C. 21%
- D. 27.59%

Correct answer: 23.81%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 23.81%.

Question 26 - Solved Example P32

Among Varad, Viraj and Akhilesh, the most consistent is

Options

- A. Varad
- B. Viraj
- C. Akhilesh
- D. all equally consistent

Correct answer: Varad

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives Varad.

Question 27 - Solved Example P32

If selection is based on the highest expected score, select

Options

- A. Viraj
- B. Varad
- C. Akhilesh
- D. none

Correct answer: Viraj

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives Viraj.

Question 28 - Solved Example P33

For company X shares with mean 50 and variance 7, S.D. is approximately

Options

- A. 2.64
- B. 7
- C. 5.28
- D. 1.90

Correct answer: 2.64

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 2.64.

Question 29 - Solved Example P33

For company X, C.V. is approximately

Options

- A. 5.28%
- B. 2.64%
- C. 7%
- D. 1.90%

Correct answer: 5.28%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 5.28%.

Question 30 - Solved Example P33

For company Y shares with mean 105 and variance 4, C.V. is approximately

Options

- A. 1.90%
- B. 4%
- C. 2.64%
- D. 5.28%

Correct answer: 1.90%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 1.90%.

Question 31 - Solved Example P33

The more stable company share is

Options

- A. company Y
- B. company X
- C. both equally stable
- D. neither

Correct answer: company Y

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives company Y.

Question 32 - Let's Construct P33

To compare word-length variability in a passage, calculate mean, S.D. and

Options

- A. coefficient of variation
- B. range only
- C. median only
- D. mode only

Correct answer: coefficient of variation

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives coefficient of variation.